

5. Sample problem: If the ocular has a 10x lens and the objective has a 45x lens the total magnification is $10 \times 45 = 450x$

Changing objectives:

1. When changing objectives from scanning power to lower power to high power the following changes will occur:
 - a. The size of the field of view decreases
 - b. The field of view becomes darker
 - c. The size of the image increases
 - d. The resolution (ability to see detail) increases
 - e. The working distance between the slide and the objective lens decreases
 - f. The depth of focus (thickness of the specimen that is visible) is reduced
2. When changing from scanning to low power the field of view gets smaller. In fact, every time you increase the power of the objective, the field gets smaller.

Steps for Using the Microscope:

1. Place the slide on the stage lining it up with the rectangle and using the stage clip to hold it in place.



2. Click the nosepiece to the lowest (shortest) setting, the **scanning objective** lens or **4x**.
3. Look into the eyepiece.
4. Use the **coarse adjustment knob** to bring the specimen into view. **The specimen must be in focus before moving to the next steps.**
5. Rotate the nosepiece to the **low-power** objective or **10x**.
6. Refocus using the coarse adjustment knob.
7. Move the slide to get a centered view.
8. Now use the fine adjustment knob to get the specimen in perfect focus.
9. Your slide **MUST** be focused on low power before attempting this next step.